

ADAUSDT Hybrid Forecasting System Model Evaluation Report

Asset: ADA

Market: ADAUSDT

Data period: 2022-08-27 to 2026-08-27

Prediction horizons: 1H, 1W, 1M

Canonical frequency: 1H

Models: ARIMA, LightGBM, AutoEncoder, AE+LightGBM, ResNet18, Fusion LightGBM

Device: cpu

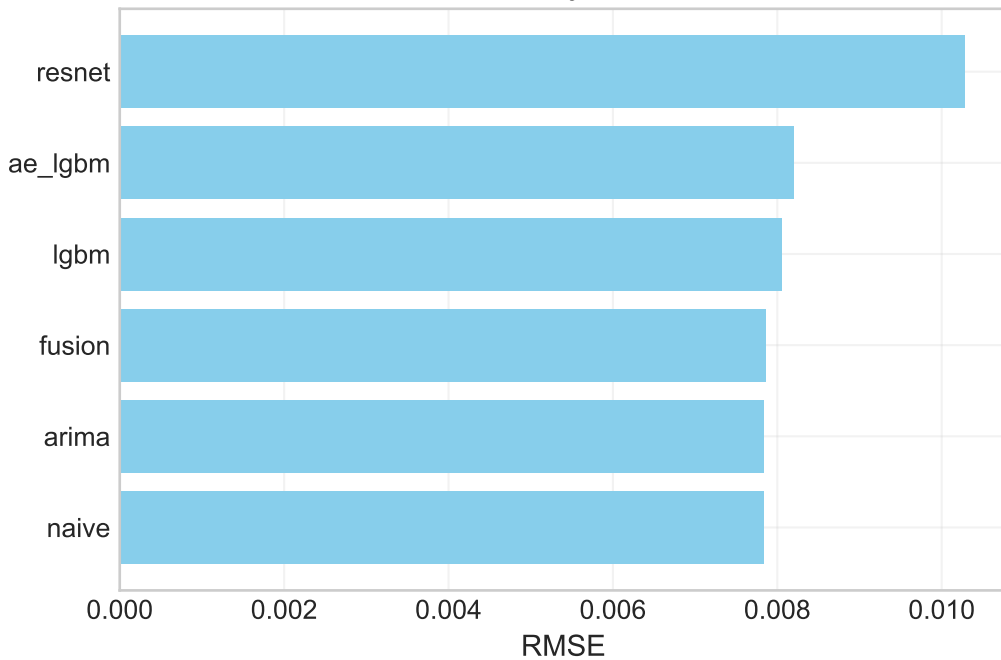
Rows (train/val/test): 24935 / 4416 / 5736

Model Performance Metrics Summary

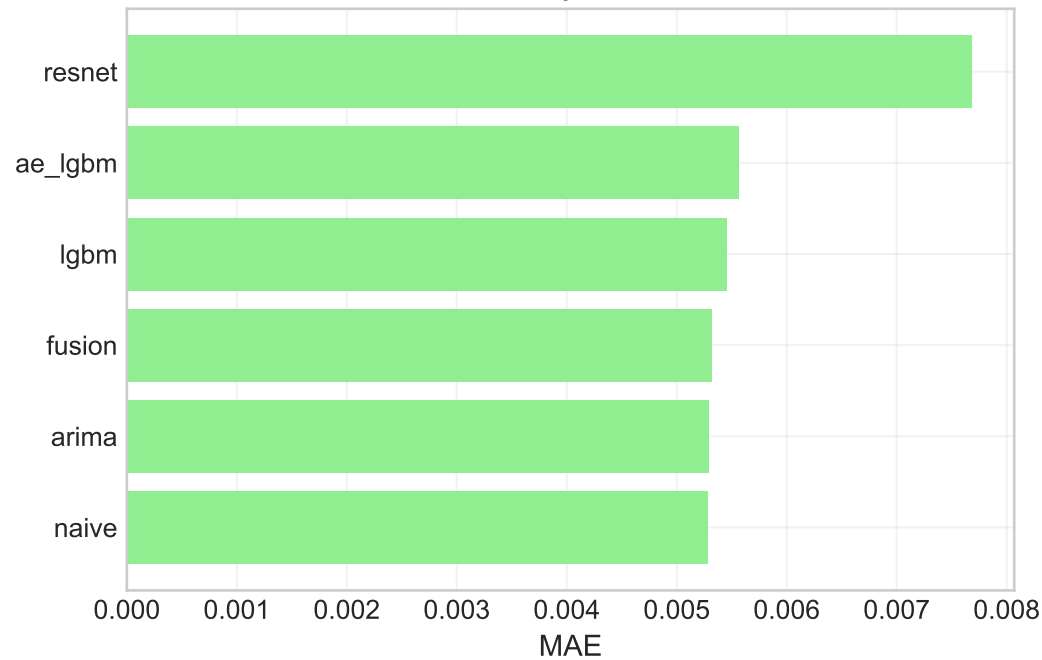
model	horizon	Directional_Accuracy	MAE	MAPE	R2	RMSE
ae_lgbm	1H	49.4159	0.0056	556086.3627	-0.0955	0.0082
ae_lgbm	1M	65.1442	0.1735	934629.2154	-0.2046	0.2265
ae_lgbm	1W	48.8147	0.1294	4336433.3828	-1.355	0.1637
arima	1H	48.3173	0.0053	80884.7514	-0.0015	0.0078
arima	1M	46.1939	0.1662	12046.0392	-0.2354	0.2294
arima	1W	47.5934	0.0795	16005.4481	-0.0264	0.108
fusion	1H	49.2066	0.0053	174063.0105	-0.0072	0.0079
fusion	1M	59.375	0.2363	3793726.6955	-0.6484	0.265
fusion	1W	44.9892	0.1138	2005469.6146	-0.763	0.1416
lgbm	1H	48.9625	0.0055	401707.327	-0.0567	0.0081
lgbm	1M	44.4111	0.3533	4832994.3742	-2.6899	0.3965
lgbm	1W	45.4203	0.1534	5612961.9162	-2.4134	0.197
naive	1H	3.4874	0.0053	96.5126	-0.0001	0.0078
naive	1M	0.1603	0.1662	99.8397	-0.2358	0.2295
naive	1W	0.2514	0.0796	99.7486	-0.027	0.1081
resnet	1H	47.6783	0.0077	1943110.7903	-0.8436	0.0103
resnet	1M	53.1509	0.1694	1156188.8072	-0.2258	0.2266
resnet	1W	57.9809	0.0756	1593063.6863	-0.0613	0.1025

Metrics for 1H Horizon

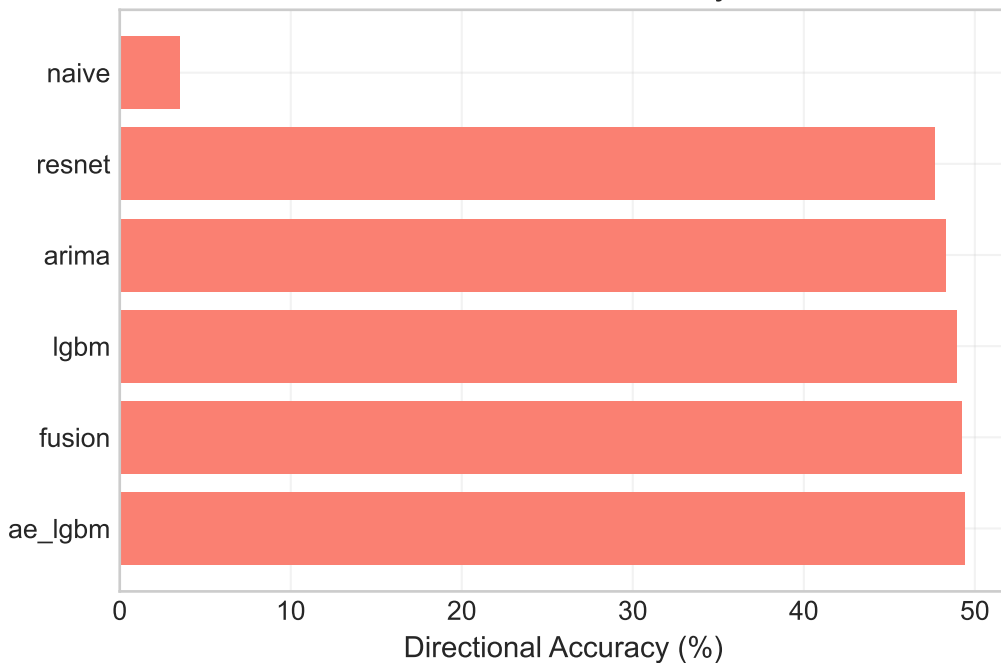
RMSE by Model



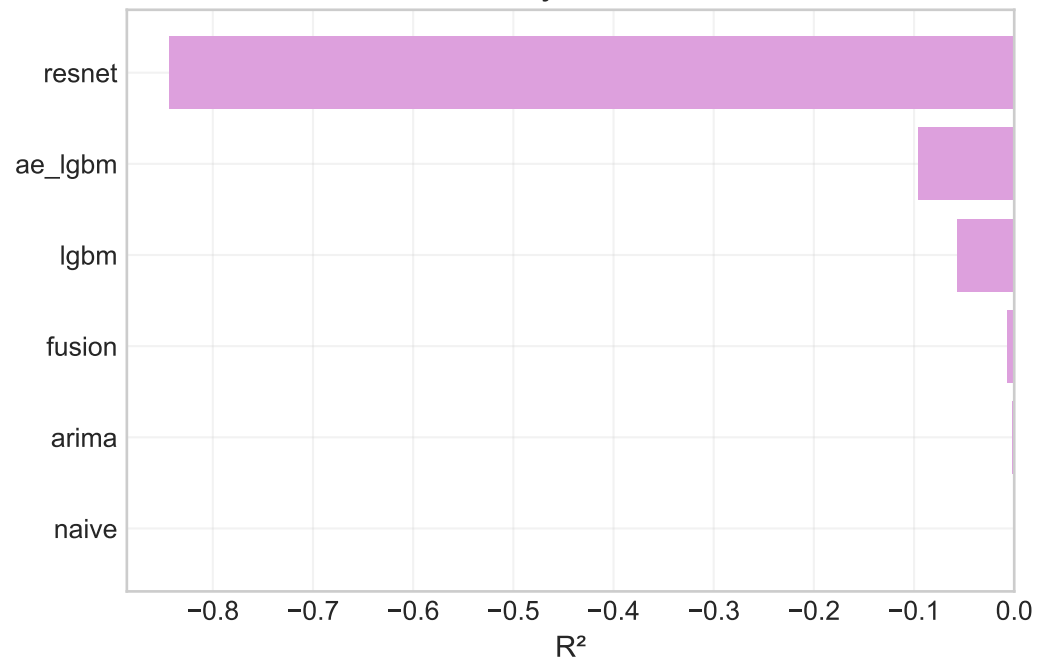
MAE by Model



Directional Accuracy

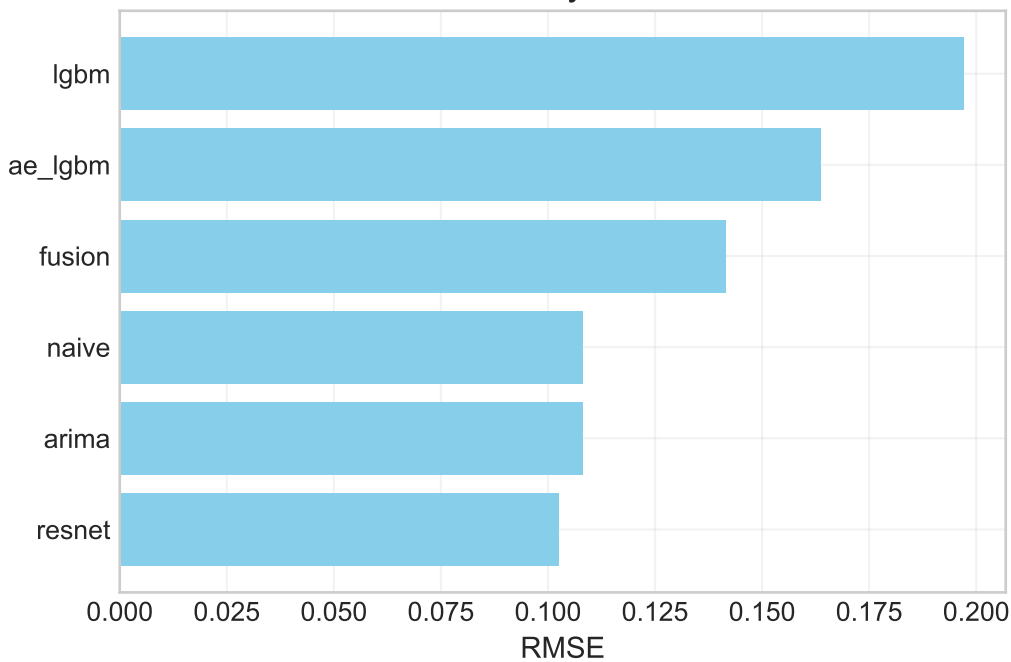


R² by Model

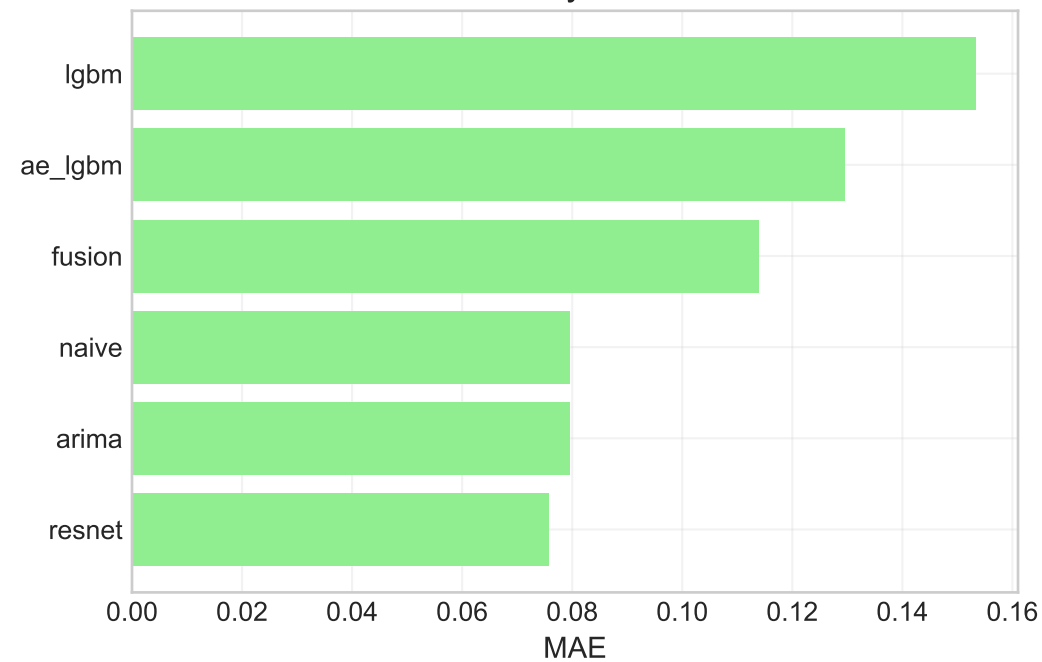


Metrics for 1W Horizon

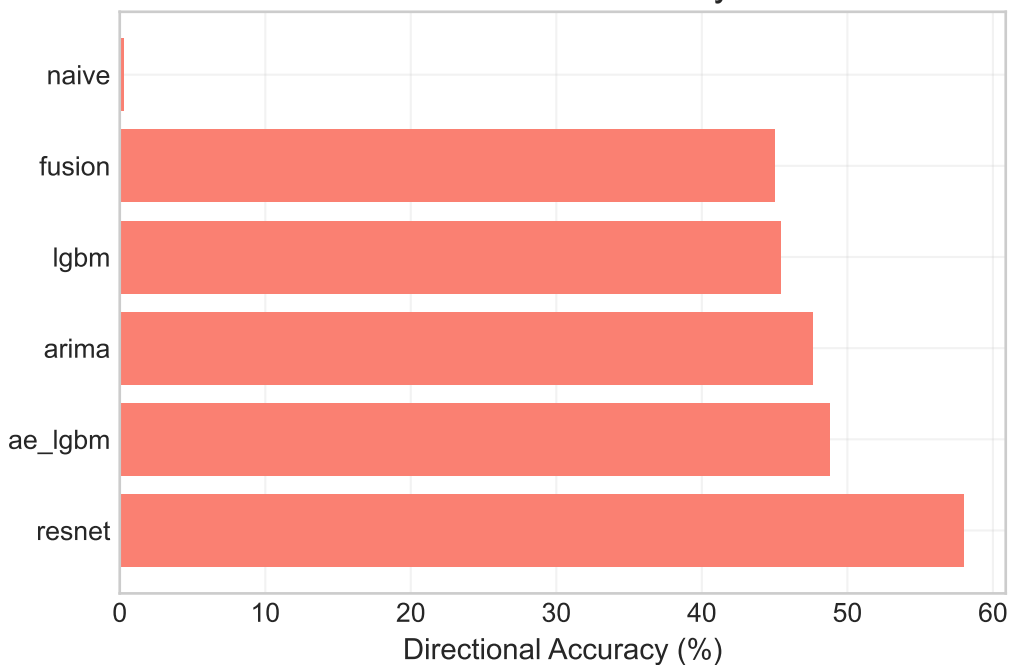
RMSE by Model



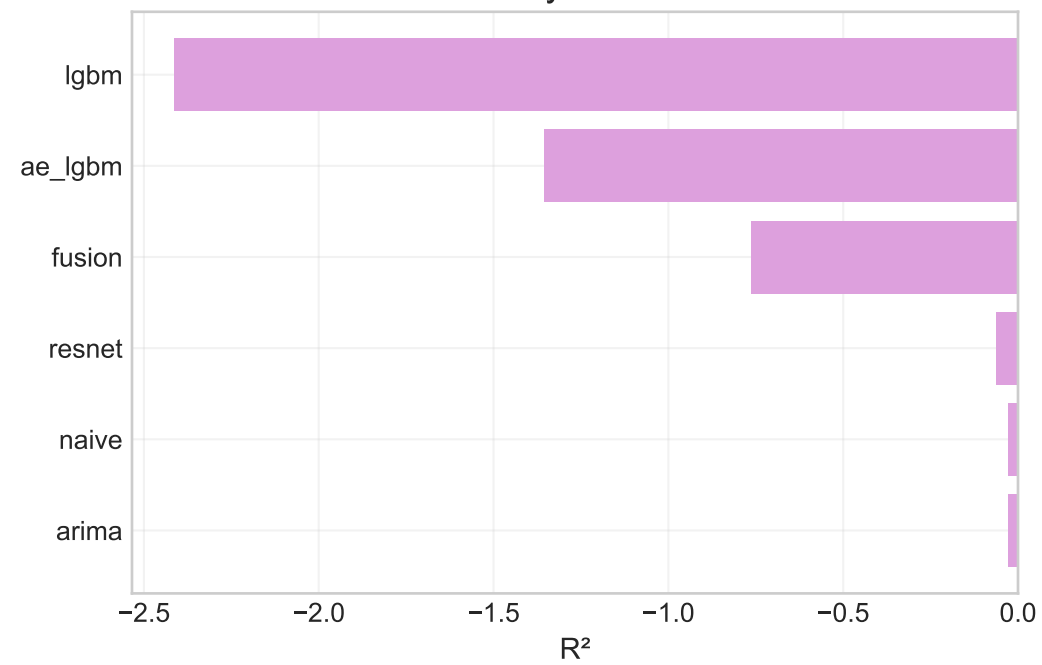
MAE by Model



Directional Accuracy

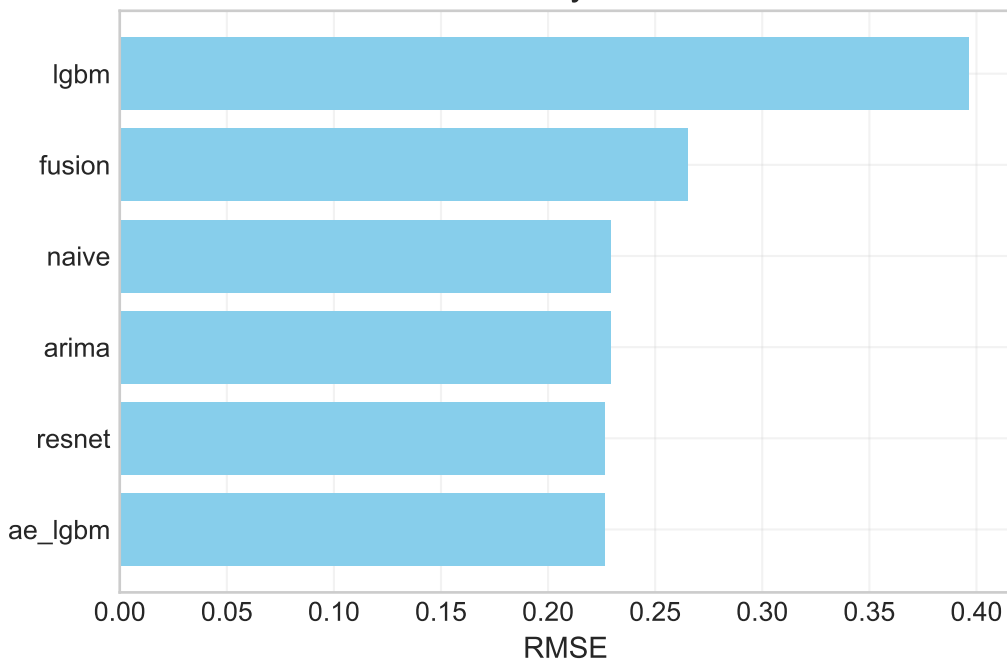


R² by Model

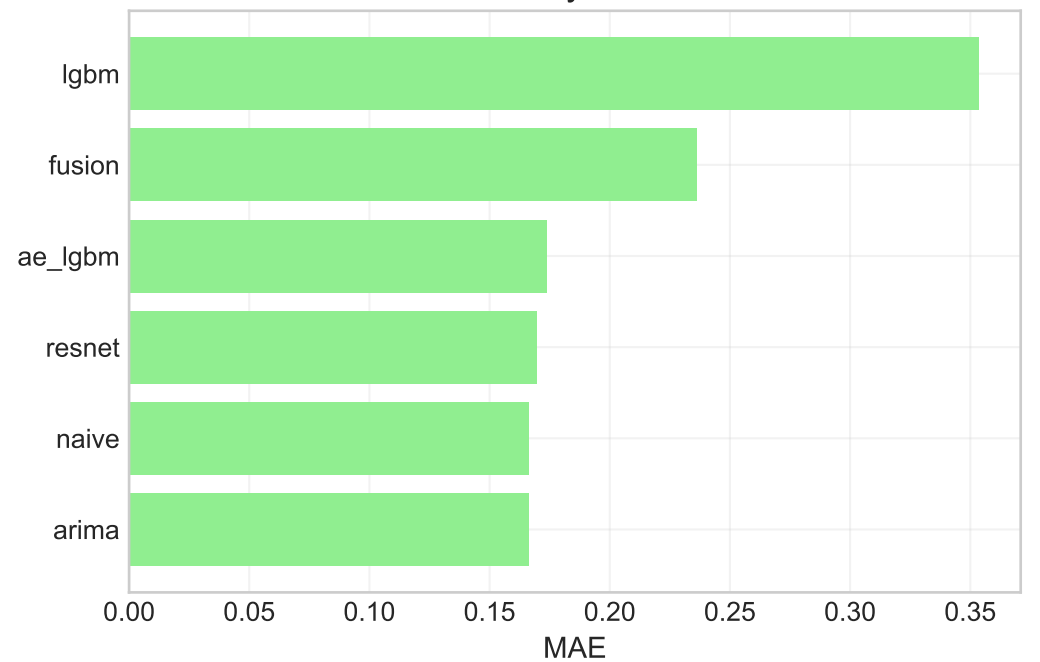


Metrics for 1M Horizon

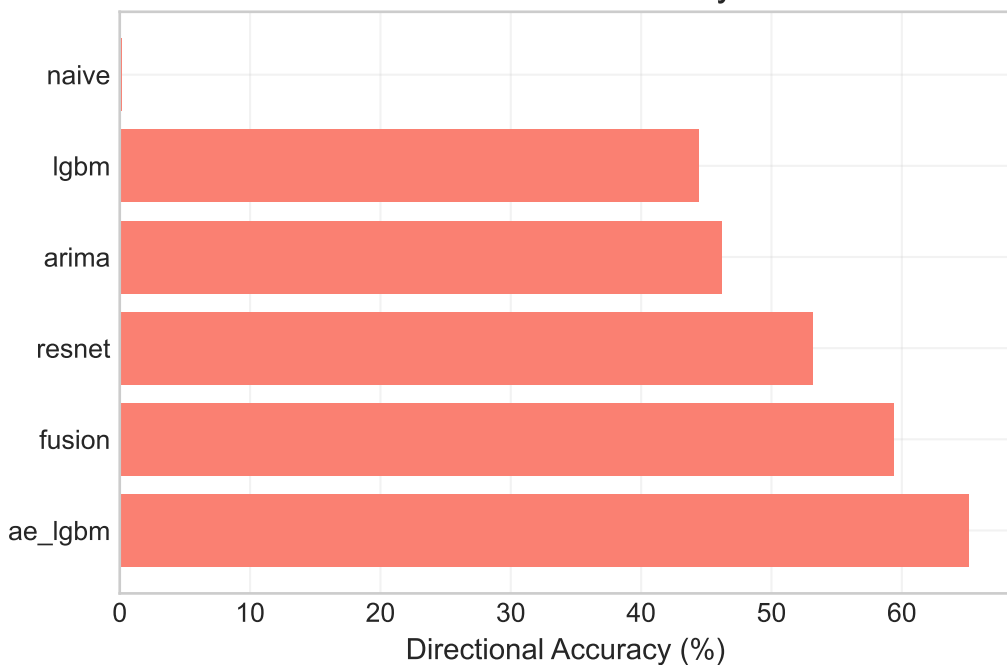
RMSE by Model



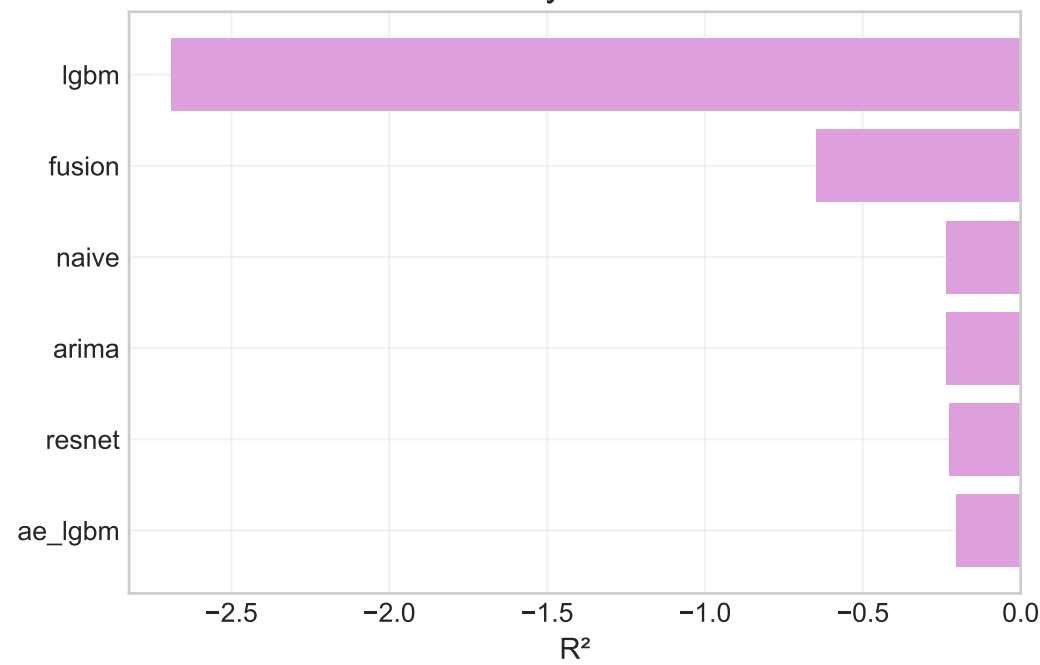
MAE by Model



Directional Accuracy

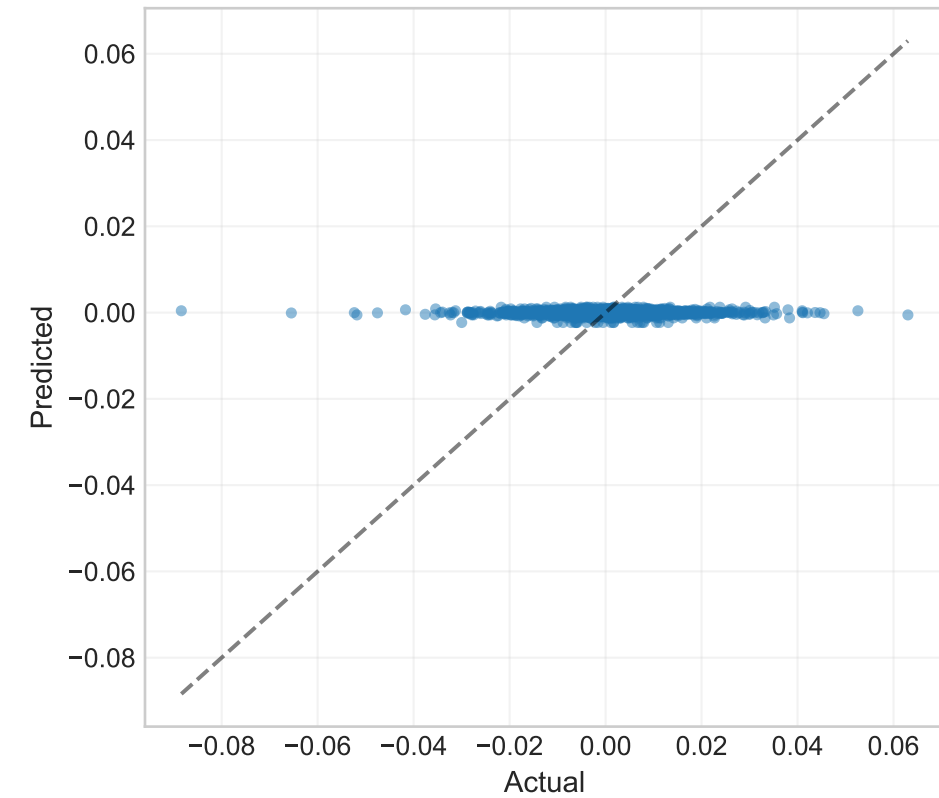


R² by Model

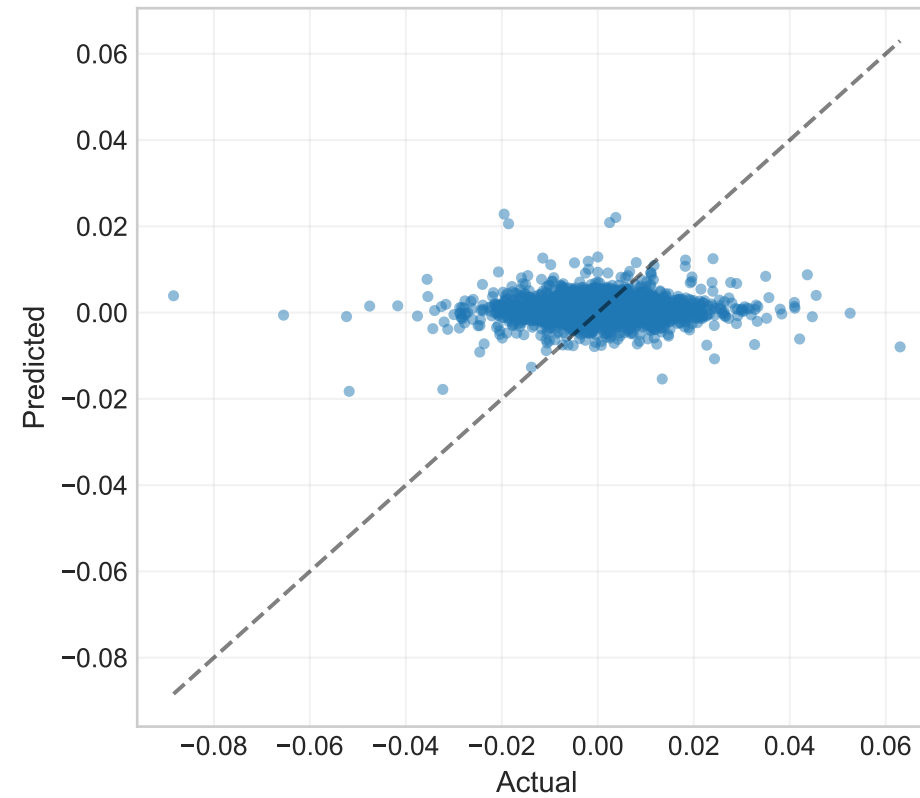


Predicted vs Actual Log Returns (1H)

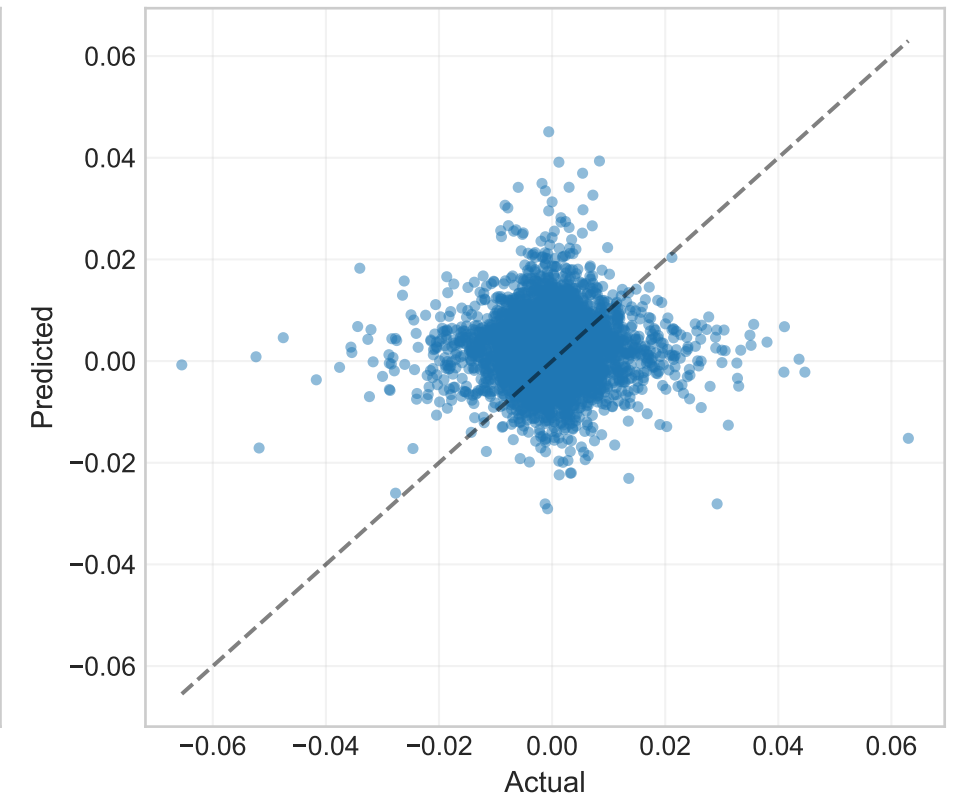
arima



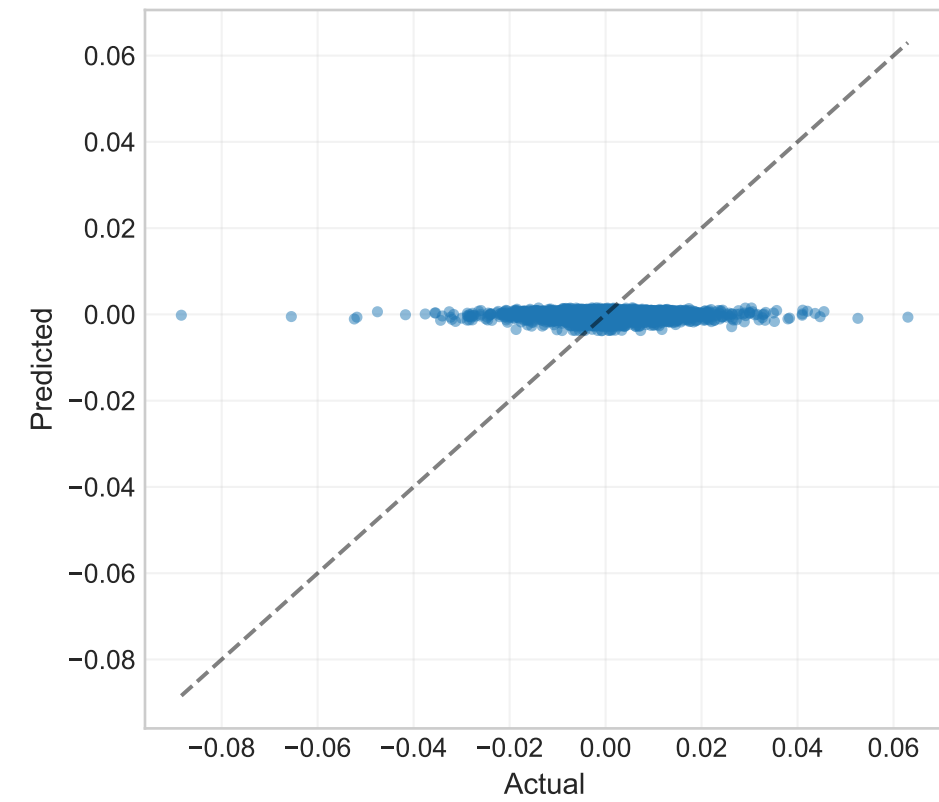
lgbm



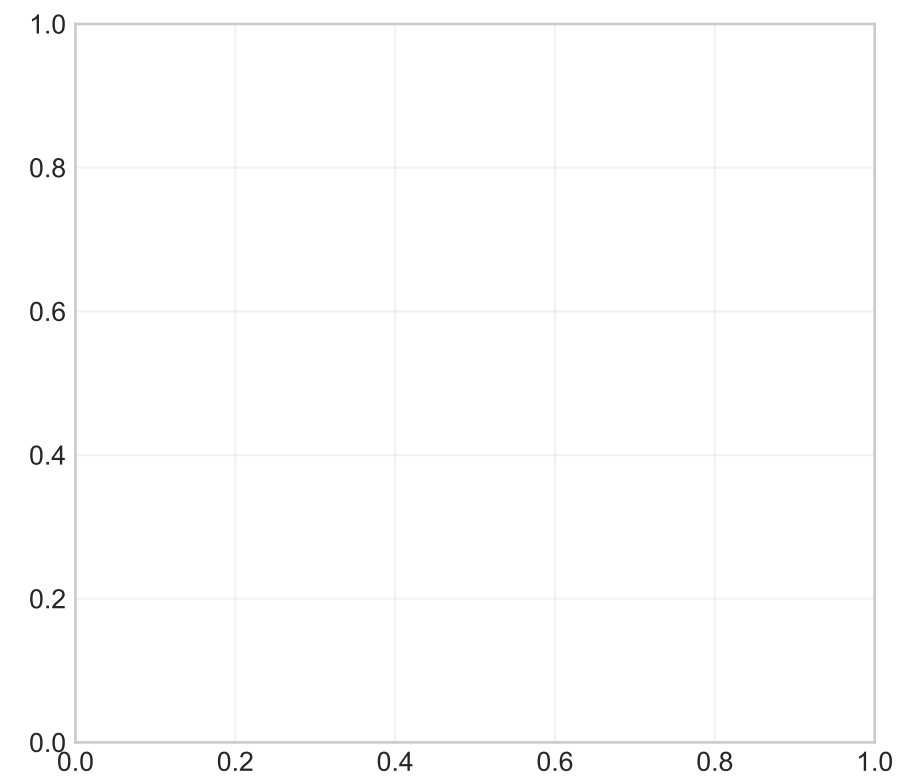
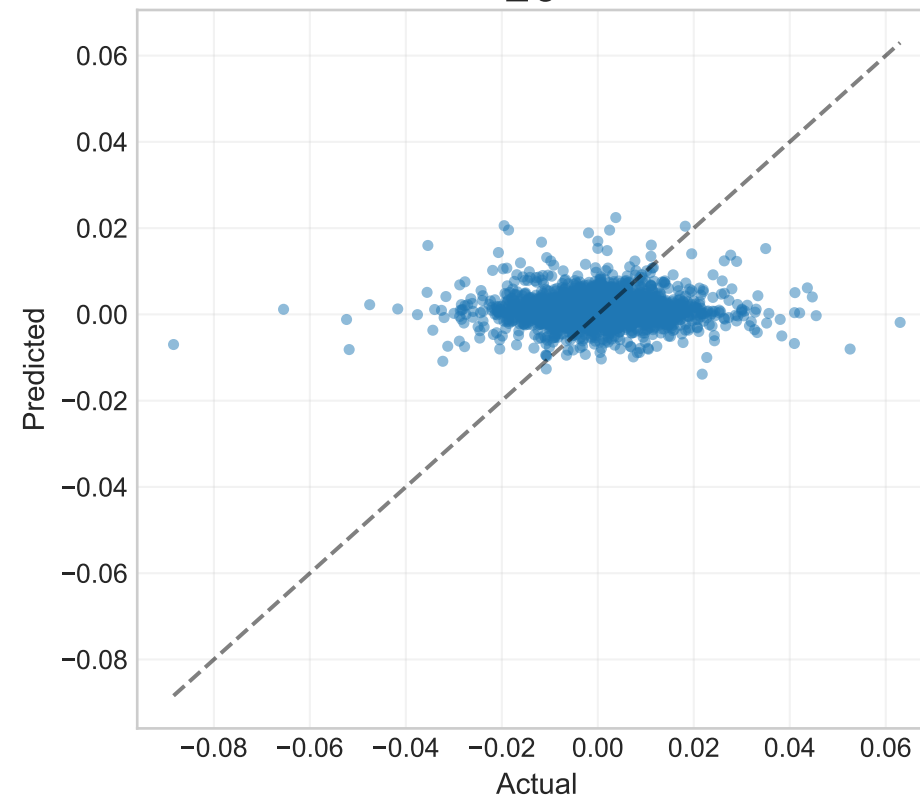
resnet



fusion

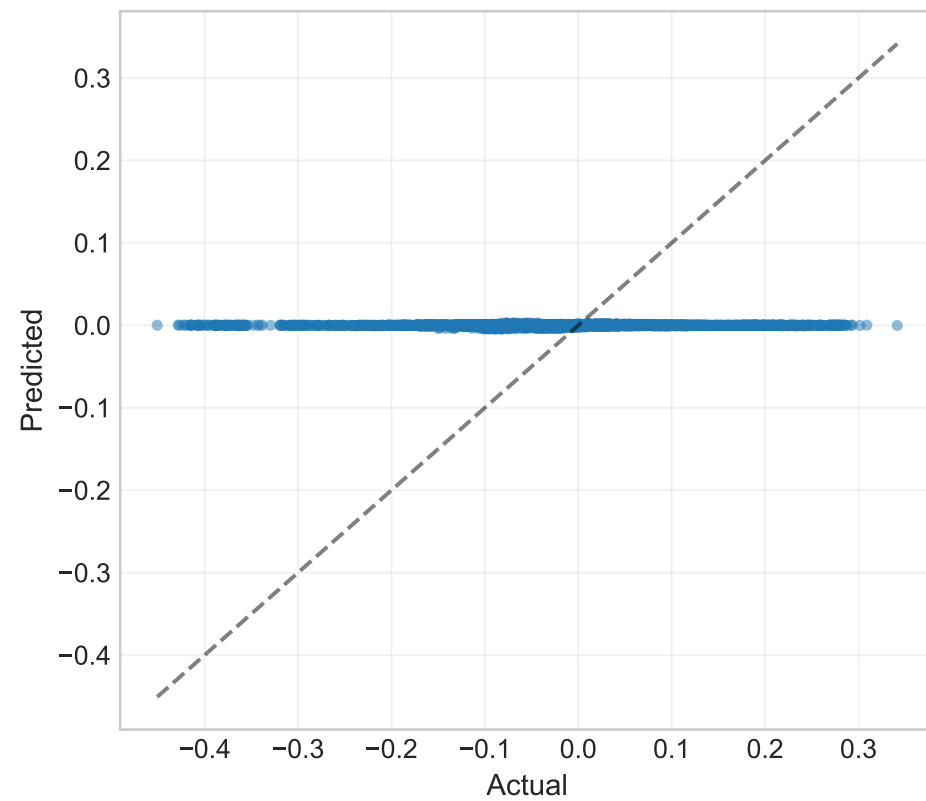


ae_lgbm

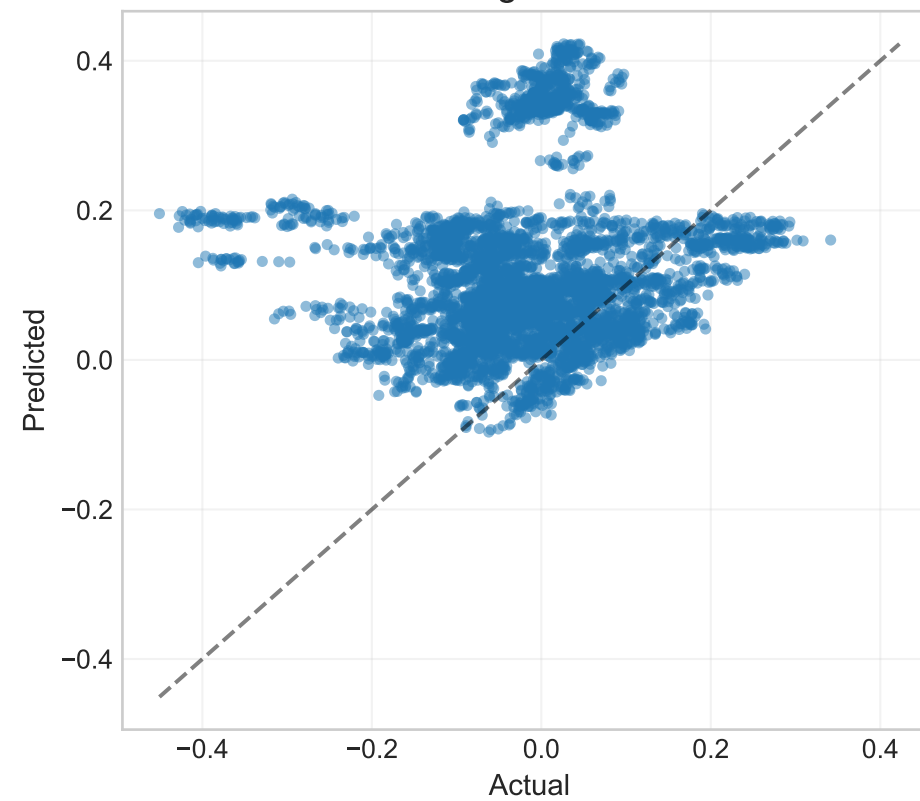


Predicted vs Actual Log Returns (1W)

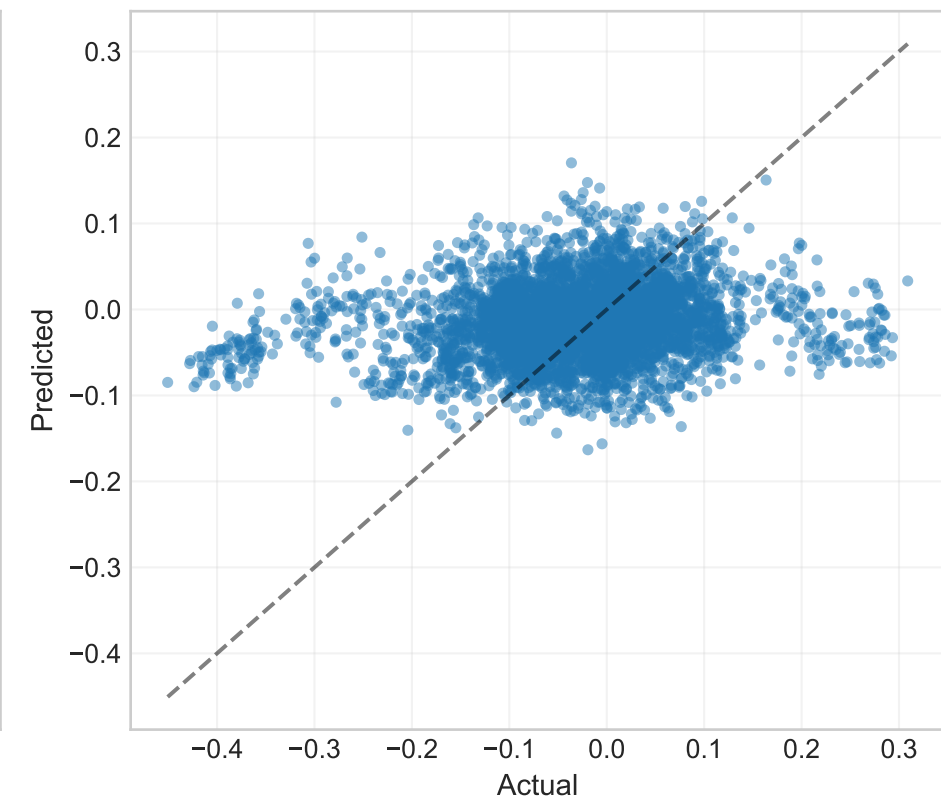
arima



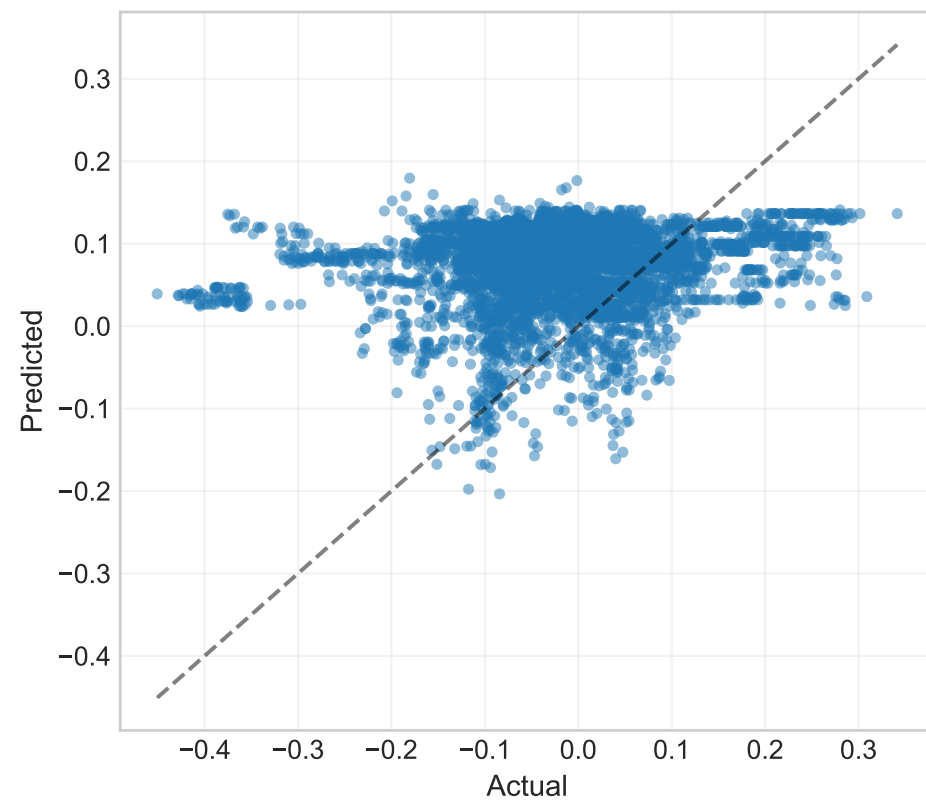
lgbm



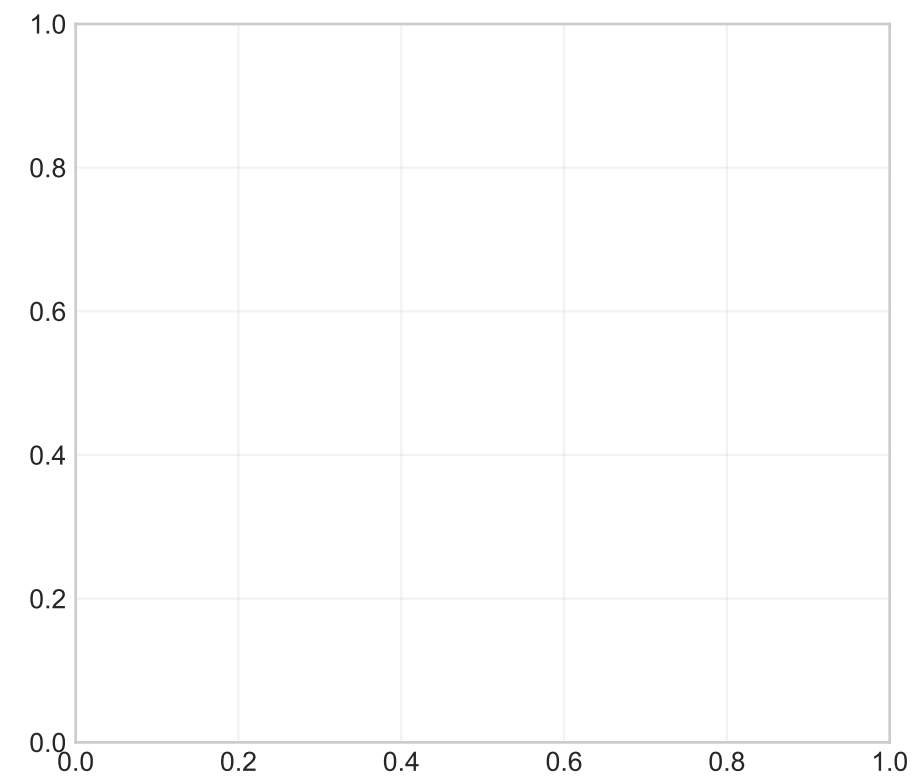
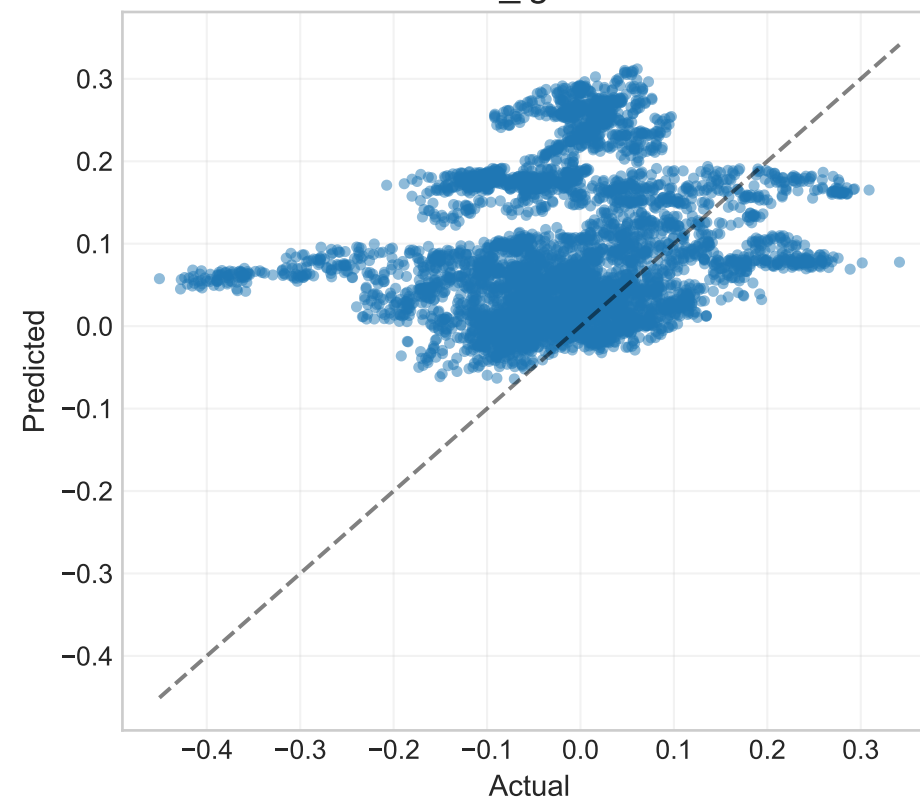
resnet



fusion

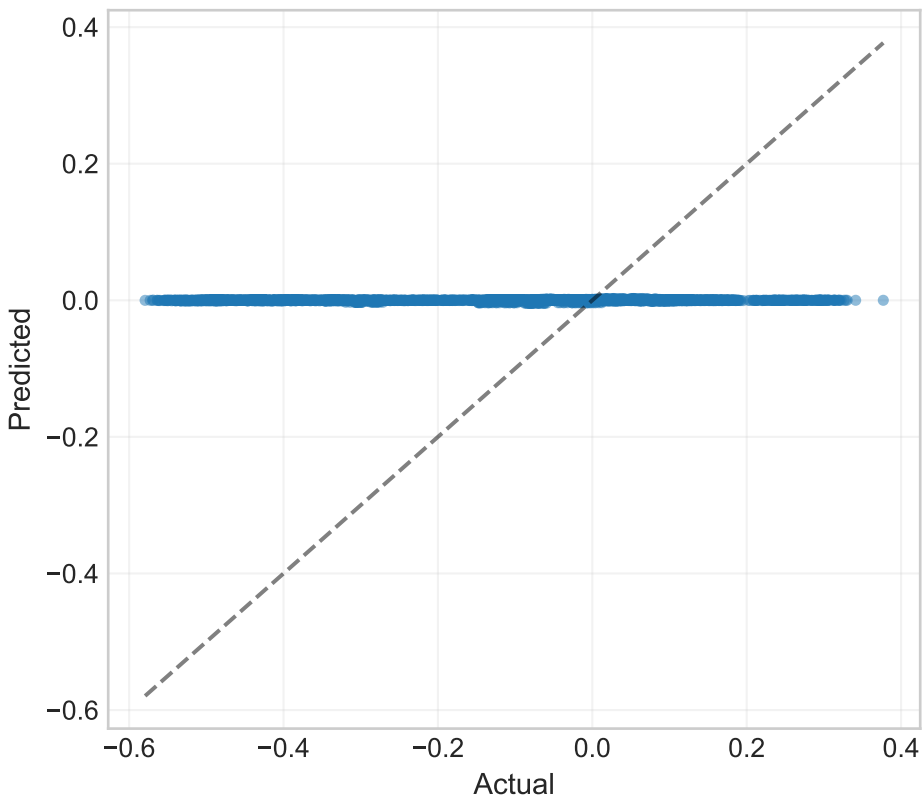


ae_lgbm

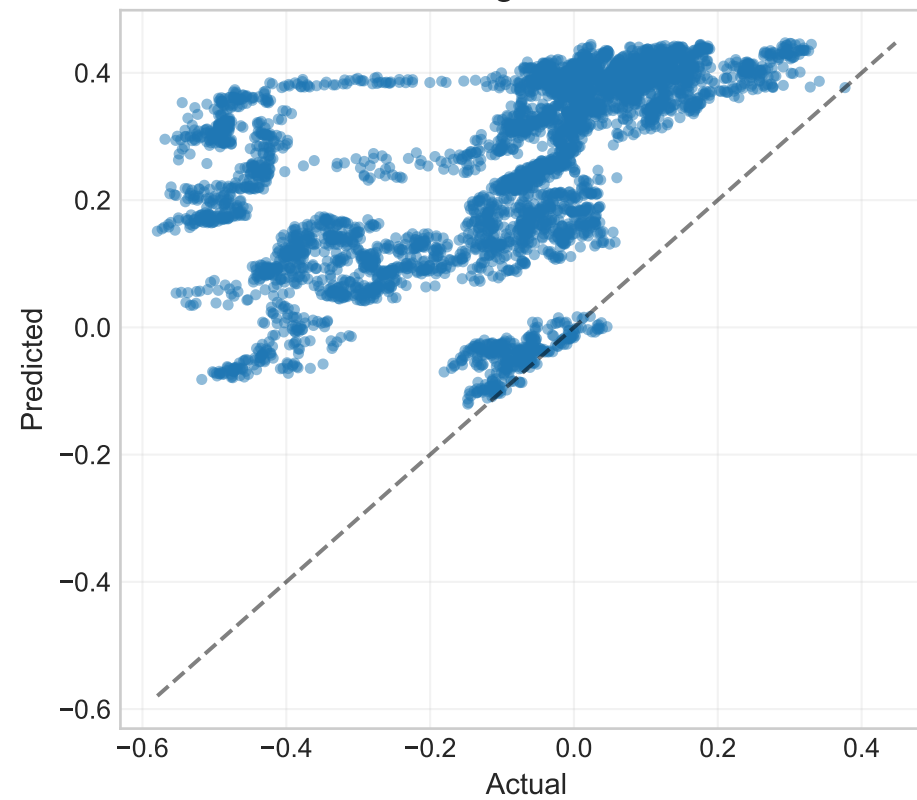


Predicted vs Actual Log Returns (1M)

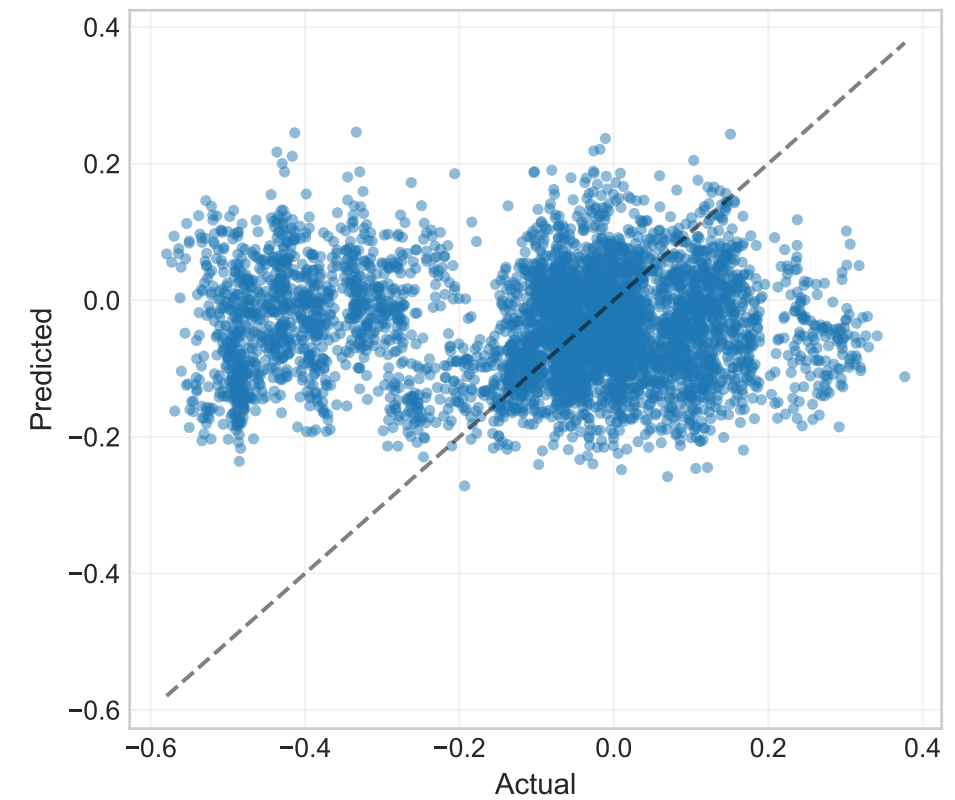
arima



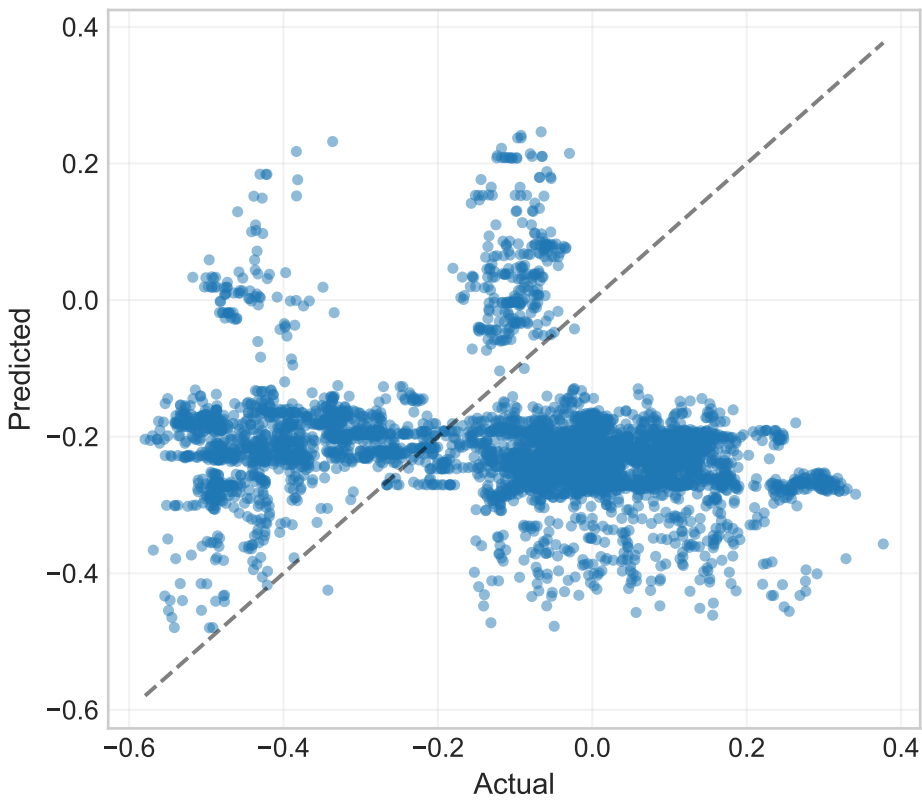
lgbm



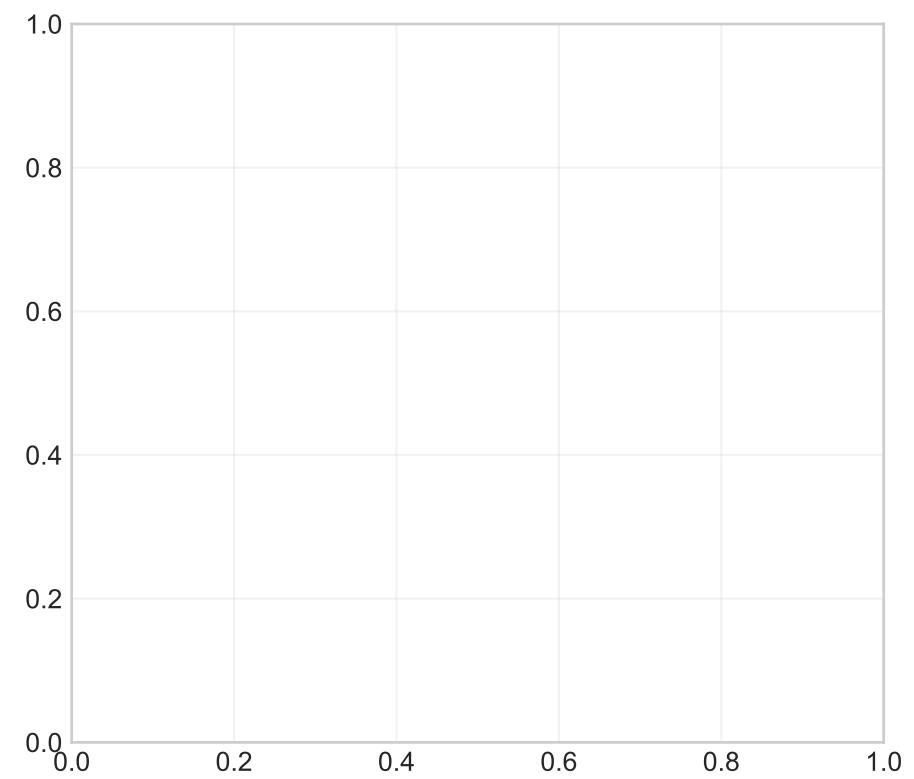
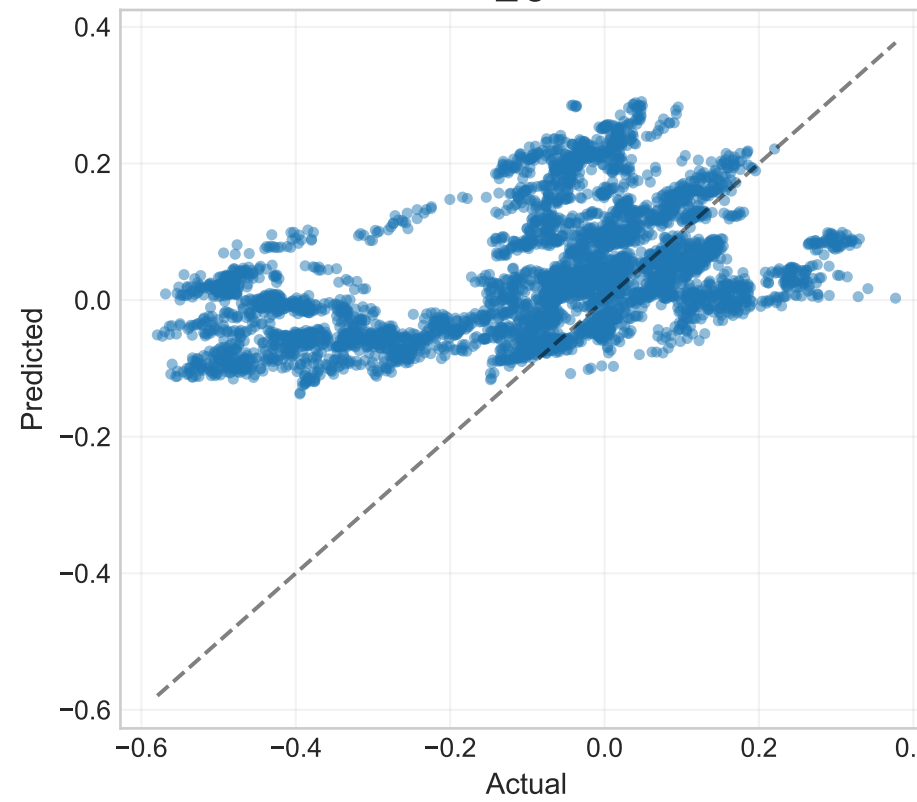
resnet



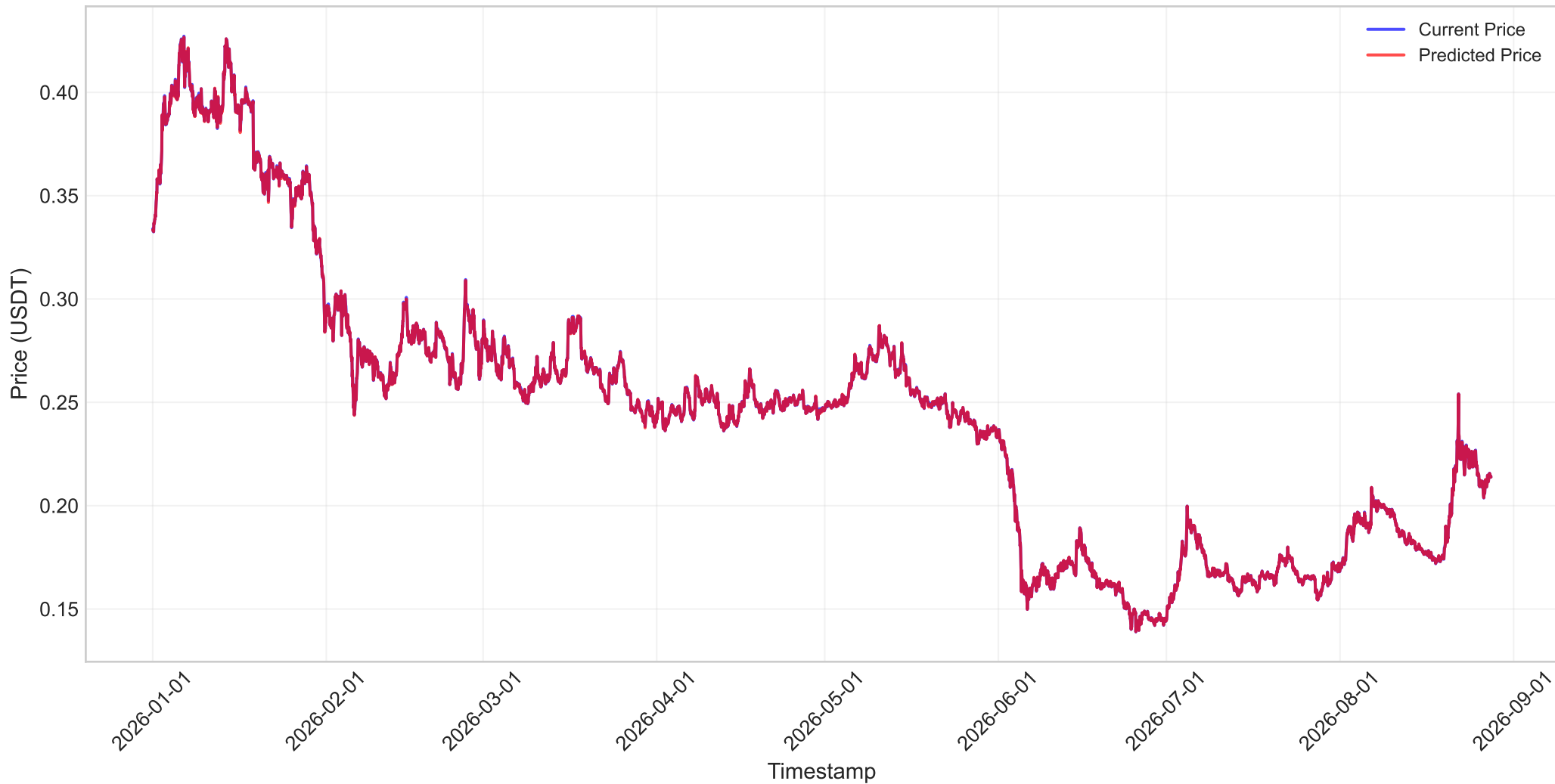
fusion



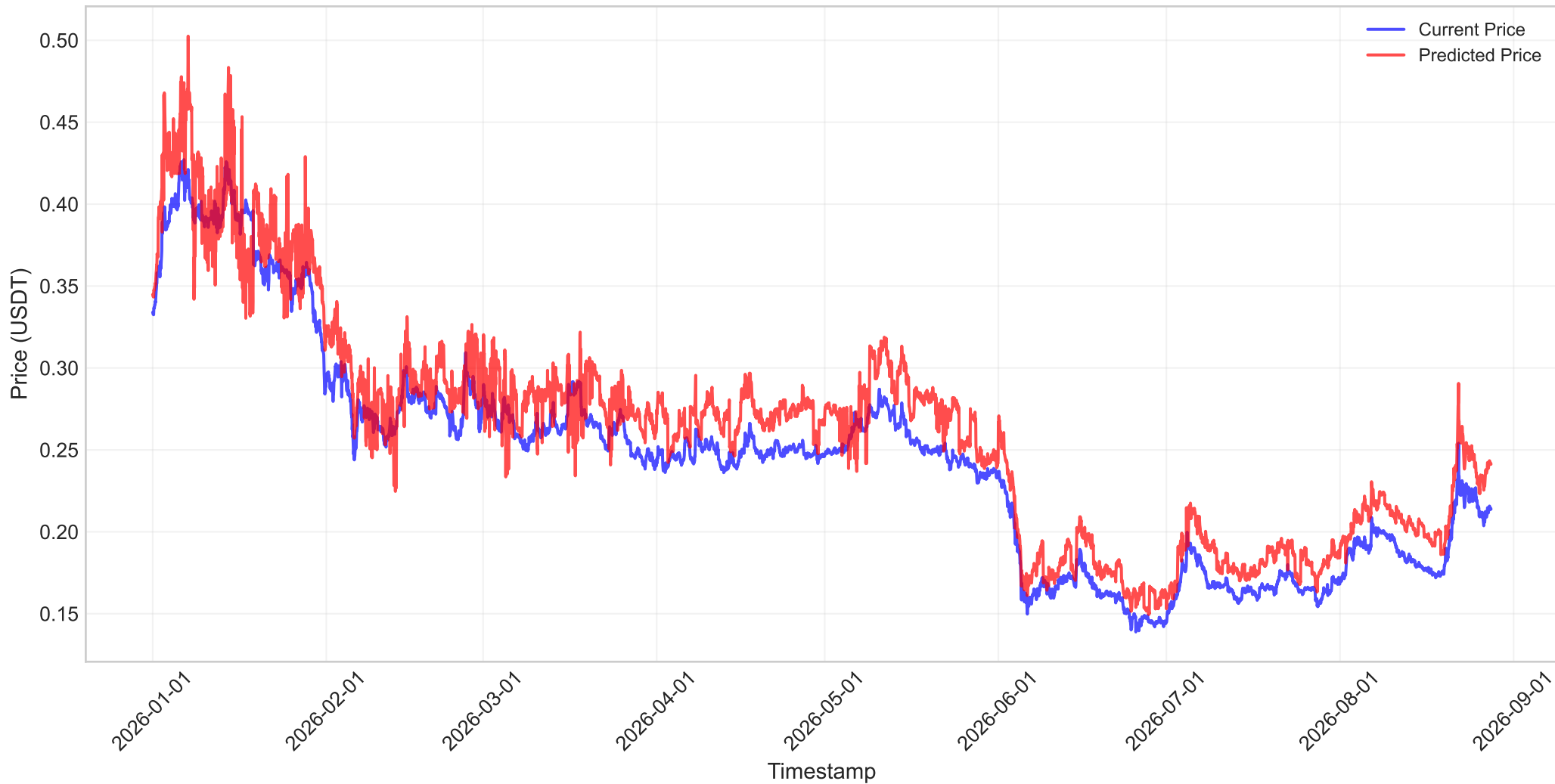
ae_lgbm



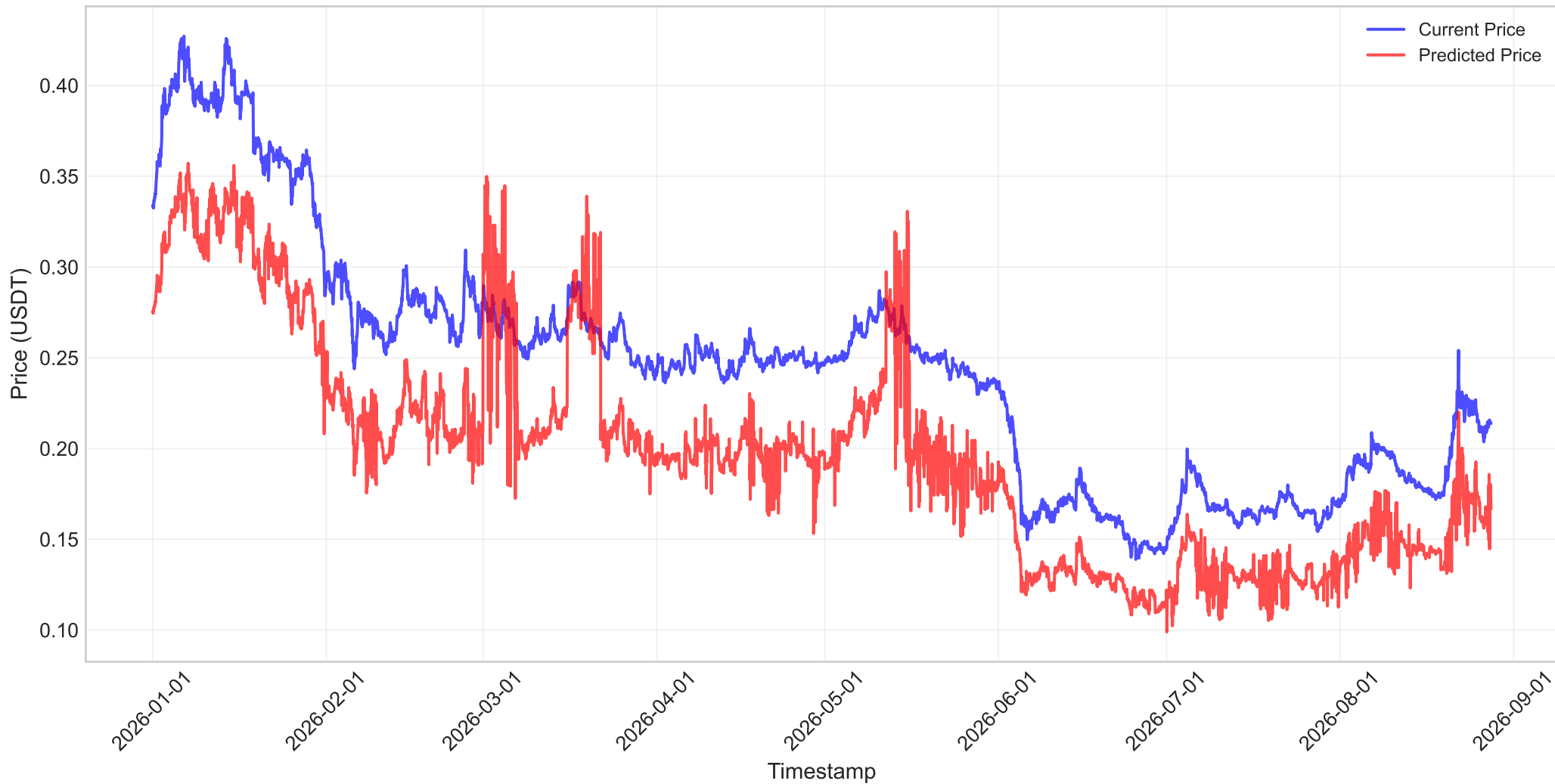
Fusion Model Predicted Price vs Current Price (1H Horizon)



Fusion Model Predicted Price vs Current Price (1W Horizon)

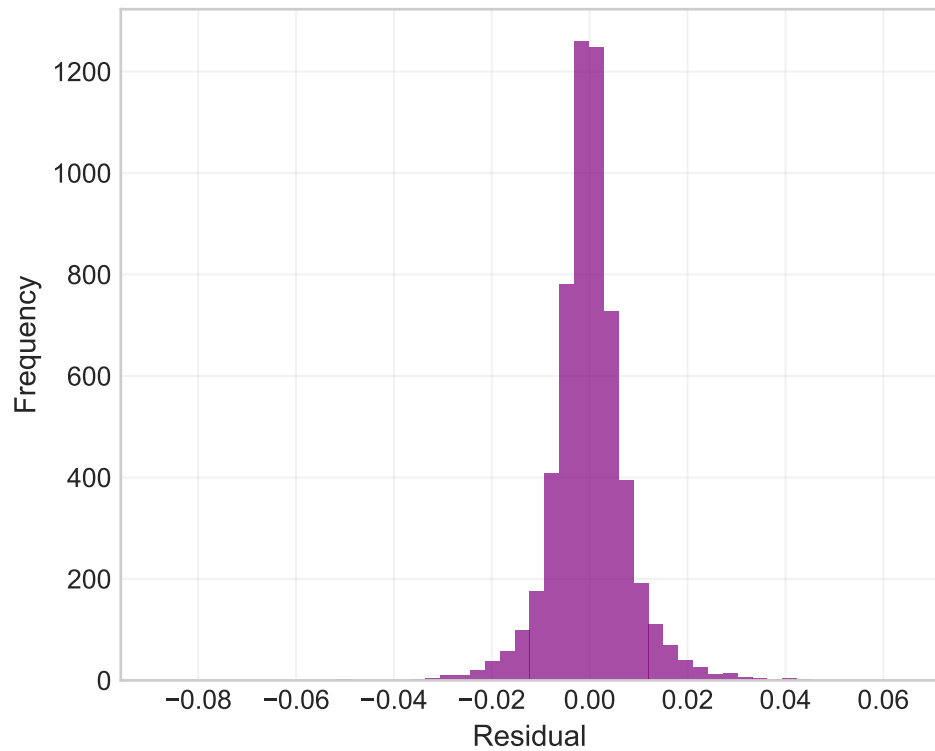


Fusion Model Predicted Price vs Current Price (1M Horizon)

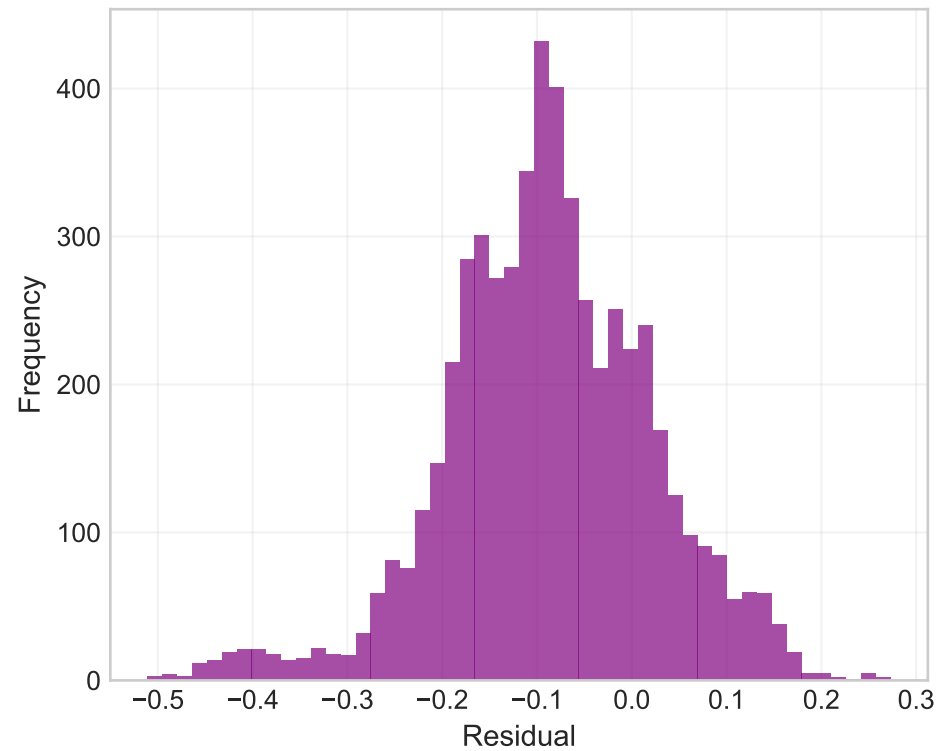


Residuals of Fusion Model (Actual - Predicted Log Returns)

1H Residuals



1W Residuals



1M Residuals

